CHAPTER 9 SOURCES AND EFFECTS OF HEAT ENERGY

In your previous class you have already learnt about transmission of heat and three different modes of heat transfer i.e. conduction, convection and radiation. In this chapter you will further explore about the sources and effects of heat energy. You already know that sun is the major natural source of heat and light during day time. Heat reaches to the earth in the form of radiation. Have you ever wondered about other natural sources of heat? Do you know why do we use artificial sources to generate heat? Have you ever thought why heat is so important for living things? What are the effects of heat in our daily life? How do solids, liquids and gases contract and expand? Let's explore these and many other related questions.

In this Chapter you will learn about:

- Sources and Effects of Heat
- Thermal Expansion and contraction (Solids, Liquids and Gases)
- Application of expansion and contraction of solids (Riveting, Fixing a Metal tyre into Wheel, Fixing Axle of a wheel, Fire Alarms and Electric Iron.)
- Effects of Expansion and Contraction of solids in everyday life. (Concrete Road Surfaces, Railway Tracks, Bridges, Overhead power and Telephone lines, Pipe lines)
- Uses of Expansion and Contraction of liquid.
- Peculiar Behaviour of Water during Contraction and Expansion

All the students will be able to:

- ➢ Describe the sources and effects of heat.
- ➢ Explain thermal expansion of solids, liquids and gases.
- ➢ Explore the effects and applications of expansion and contraction of solids.
- ➢ Describe the uses of expansion and contraction of liquids.
- ➢ Explain the peculiar behaviour of water during contraction and expansion.
- ➢ Investigate the processes making use of thermal expansion of substance.
- ➢ Identify the damages caused by expansion and contraction in their surrounding and suggest ways to reduce the damages.
- ➢ Investigate the means used by scientist and engineers to overcome the problems of expansion and contraction in everyday life.
- ➢ Describe the working of a thermometer.

Fig. 9.1 Sources of Heat Energy

Fig. 9.2 Thermal expansion in Solid

SOURCES AND EFFECTS OF HEAT

> ➢ Describe the sources and effects of Heat

Heat is a form of energy found due to the random motion or vibration of atoms, molecules and ions. You have already learnt that heat energy is the capacity for action or performing work and it flows from a region of higher temperature to the region of lower temperature. It means heat flows from things that are hot. Sun is the natural source of heat energy, while artificial sources are wood, coal, electricity, oil and gas.

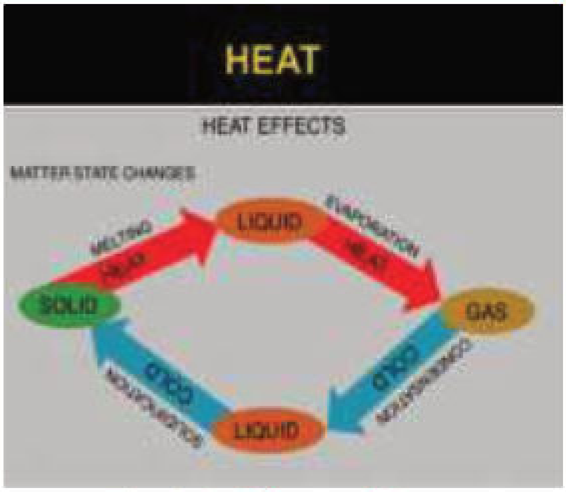


Fig. 9.3: Effects of Heat

Heat energy brings out chemical changes in a substance, for example, when marble $CaCO_3$ (calcium carbonate) is heated, it turns into calcium oxide (CaO) and carbon dioxide (CO_2). Also, a body may catch fire if it is sufficiently heated. The burning of substance in air with the release of large amount of heat and light energy is called combustion.

THERMAL EXPANSION AND CONTRACTION (SOLIDS, LIQUIDS AND GASES)

> ➢ Explain thermal expansion of solids, liquids and gases.

Thermal expansion of solids, liquids and gases is the increase of the size (length, area and volume) of a body due to a change in temperature. While decrease in size (length area and volume) of a body due to change in temperature is called thermal contraction.

All the three states of matter solids, liquids and gases expands upon heating and contract upon cooling. Thermal expansion is large for gases and relatively small for liquids and solids. Let us explore the effects and application of thermal expansion, contraction, and its effects and applications in solids.

- Explore the effects and applications of expansion and contraction of solids.

a) Thermal Expansion of Solids

Fig 9.4 Electrical Wires are left slack

You have learnt in your previous class that material objects solids, liquids and gases are made up of tiny particles; atoms and molecules. In solids, particles are closely packed with each other. When solids are heated the vibratory motion of their particles (atoms and molecules) become fast and they begin to push each other further apart, thus results into expansion of solids. Similarly, when solids are cooled particles slow down, come closer to each other and solids contract. The expansion and contraction caused by heat is also known as thermal expansion or contraction respectively. It means heat energy or thermal energy can change the length of solids and volume of liquids and gases. You may have noticed that the telephone and electric wires are not hung tightly, and left slack. Why? Also, these wires are loosened during hot summer season, why? What change in the length of wires have you observed during winter season?

Wires are left slack so that they are free to change length. Let us perform a simple activity to understand this phenomenon easily through experiment.

Activity 9.1:

Exploring thermal expansion in solids.

What I need?

- One-meter long copper wire
- Iron stand to be used to suspend the fully stretched copper wire
- Candle/sprit lamp
- Match box

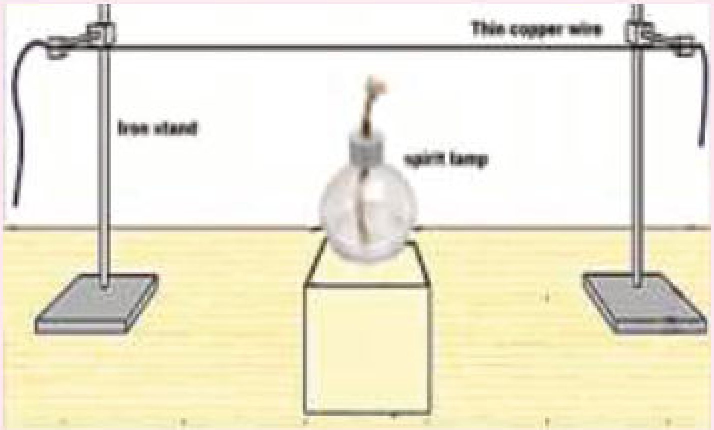


Fig. 9.5 Experiment showing heating the copper wire with spirit lamp

What to do?

1. Wind a one-meter long copper wire around the two iron stands.
2. Wire should be suspended in fully stretched position as shown in Figure 9.5.
3. Place the candle/spirit lamp under the wire in the middle.
4. Heat the wire with candle/spirit lamp. Do not touch the wire after heating.
5. Heat a few minutes. What happened to the length of wire? Record your observation and its reason.

What I Observed?

Activity Questions:

1. Why does the wire loosen and sink after heating?
2. What does the heat cause to the molecules of copper wire?
3. What happened to the length of the wire upon heating?
4. What happened to the length of wire after cooling?

What I Concluded?

- Describe the uses of expansion and contraction of liquids

b) **Thermal expansion of liquids**

Have you ever noticed that the liquid mercury in the thermometer rises on heating and falls on cooling? Let us explore.

In a liquid, expansion occurs when heated. The particles move faster around each other and expand. An example of expansion in a liquid is ocean. In hot climate the water expands, and the sea level rises due to the heat of the sun besides the hot weather. Contraction happens in liquids upon cooling.

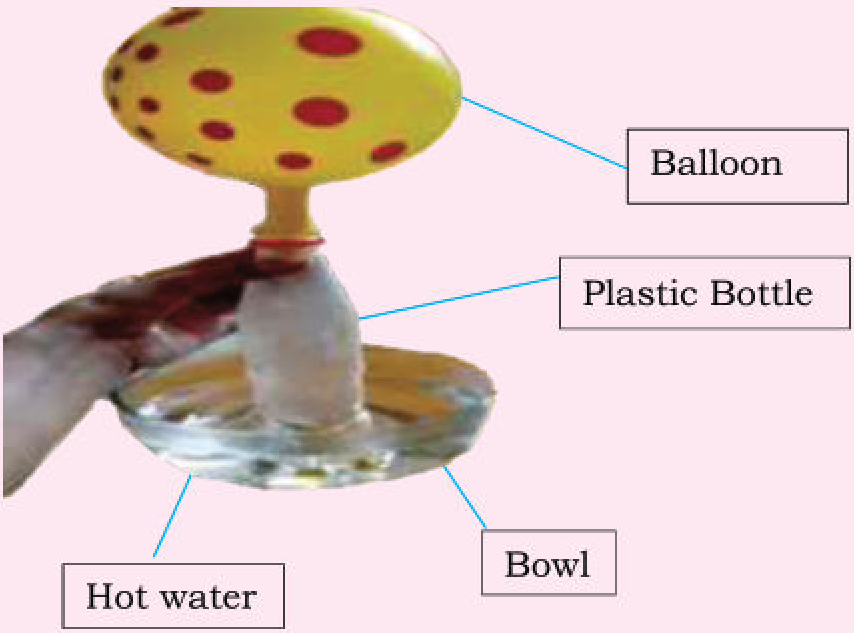
c) **Thermal expansion of gases**

Activity 9.2:

Exploring thermal expansion of air (Mixture of gases)

What I need?

1. Two bowls / water trough
2. One litre plastic bottle
3. One balloon
4. Hot water from the tap (do not use boiling water)
5. Cold water with ice cubes



What to do?

1. Take some hot water from the tap in one bowl (if hot tap water is not available then heat the water on stove in a kettle in the supervision of teacher).
2. Take cold water with ice cubes in the other bowl.
3. Blow the balloon up to stretch so that the balloon would become more flexible. Place this balloon over the mouth of the one litre plastic bottle (do remember that this bottle is not empty; it has air which is the mixture of gases).
4. Predict, what will happen when you will place this bottle with balloon in hot water and then in cold water. Discuss all responses.
5. Now place the bottle in the centre of the bowl filled with hot water. Wait a few minutes observe and see whether your predictions were right or wrong.
6. Record your observations.
7. Remove the bottle from the hot water bowl and place it in the bowl containing cold water and ice.
8. Wait for few moments and observe the balloon. Record your observation.

What I observed:

When bottle was placed in hot water:

____

When bottle was placed in a bowl containing cold water and ice:

____

ACTIVITY QUESTIONS:

1. Why it is necessary to blow and stretch the balloon before placing it on mouth of the bottle.
2. Why the balloon filled with air was inflated upon placing the bottle in hot water?

3. Why the balloon was deflated when you placed the bottle in cold water?
4. Explain and draw the movement of particles in the bottle when it was;

   a. In hot water bowl   b. in cold water bowl

What I Concluded:

DO YOU KNOW?

In the above experiment, the heat from hot water was sufficient to expand the air present in the bottle considerably. Solids, however, expand much less than gases.

All three states of matter expand upon heating because particles absorb heat and move further apart and hence, take more space. While upon cooling particles come closer and hence, it contracts and gets smaller. The initial binding forces that had kept the particles bound, now become inadequate to maintain the same form or structure. As a result, the drifting of particles causes expansion. In contrast, cooling down of particles leads to more condensation of the material, lesser movement of particles and thus lead to contraction of liquid.

If the gases are heated in a closed container, the particles collide with the sides of the container and cause pressure. When the number of collisions increases, the pressure also increases. According to the particle theory of gases, when particles are heated they move faster. As a result, gas occupies more space it is called expansion.

THE PECULIAR BEHAVIOUR OF WATER DURING EXPANSION AND CONTRACTION

✓ Explain the peculiar behaviour of water during expansion and contraction.

Fig. 9.5 Frozen water (Ice berg) in sea

Fig. 9.6 Fish and aquatic plants live under frozen water during winter season

When the temperature increases or decreases, the water behaves quite differently from other liquids. On cooling from $4^{\circ}C$ to $0^{\circ}C$ water solidifies (freezes) as ice, its volume increases, and density decreases. As a result, ice floats on the top of liquid water. This property of water helps aquatic animals and plant to survive in cold countries during winter season. Ice floats on the water surface and fish and other animals live underneath frozen lakes and ponds.

APPLICATION OF EXPANSION AND CONTRACTION OF SOLIDS

➢ Investigate the processes making use of thermal expansion of substance.

Mostly solids expand (increase in volume) when they are heated and contract (decreased in volume) when they are cooled. It means change in shape, area and volume occurs due to heating or cooling. Thermal expansion, followed by contraction upon cooling, is used in solids in the following processes:

1. **Riveting:** Rivet is a steel bolt used as permanent mechanical fastener. A rivet consists of a smooth cylindrical shaft with a head on one end. The end opposite to the head is called the tail. Before installation, a rivet is heated over a very strong flame. On installation the rivet is placed in a punched or predrilled hole, and the tail is deformed or flattened with a hammer, so that it expands to about 1.5 times the original shaft diameter. This fixes the rivet in place. When the rivet cools down, it contracts and holds the two metal plates tightly together. Rivets can fasten hard material such as wood, metal, and plastic. Rivets are commonly used in home building, wall and ceiling decorations and signs, wood working, Jewellery and air crafts.

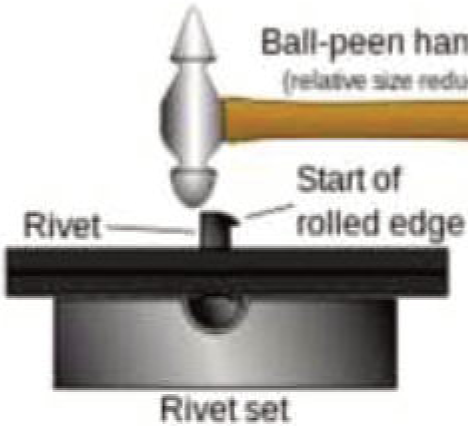


Fig. 9.7: Rivet

2. **Fixing a metal tyre onto a wheel:** The cool metal tyre is too small to fit into the wheel; therefore, it is heated. When the metal tyre is heated it expands and wheel can then fit in it loosely. However, upon cooling the tyre contracts and fits on the wheel tightly.

Fig.9.8: Metal tyre fixed into wooden wheel after heating

3. Fixing axel of a wheel: Mostly wheels of trains are fixed in axels by this method. As you have experienced that metal contracts upon cooling, this property of metals is used in this method. The diameter of axel is kept slightly larger than the hub of the metal wheel. Therefore, in order to contract it, it is placed in liquid nitrogen at -190°C. The axel cools and contracts or shrinks until it can be fitted into the hub of the wheel. Then at room temperature, it expands and fits into the wheel tightly.

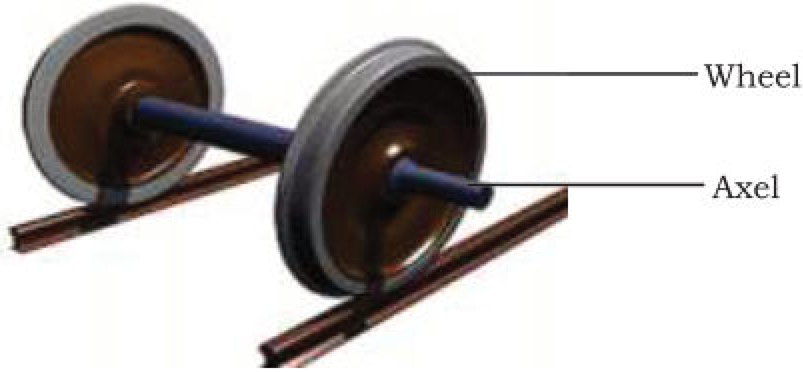


Fig. 9.9 Axel of a wheel

4. Applications of Bimetallic Strips: Bimetallic strips are used in thermostats of electrical appliances to control the temperature. Bimetallic strip is made up of two metal strips joined together; usually one strip is of steel and the other is of brass. One metal strip of bimetallic strip expands much more than the other upon heating. At room temperature the strip is flat. When heated the strip curves because the brass expands more than steel. This causes the strip to bend towards steel side. The bimetallic strip can be used as a switch to close or open a circuit. It is used in thermostats. Thermostats keep temperature constant in appliances such as electric irons, heaters, oven, fire alarms, air conditioners, car thermostat and refrigerators.

Fig. 9.10 Bimatellic Strip

Fig. 9.11 Bimetallic strip after heating

a) Electric Iron: Thermostat in an electric iron, controls the temperature of iron. When electric current flows through its heating element, it becomes hot. Bimetallic strip connected with the heating element through a spring also heats up. As a result, bimetallic strip bends and is disconnected from the heating element. This makes the circuit open and switches OFF the electric iron. On cooling the bimetallic strip straightens. The circuit is again closed, and the iron is switched ON.

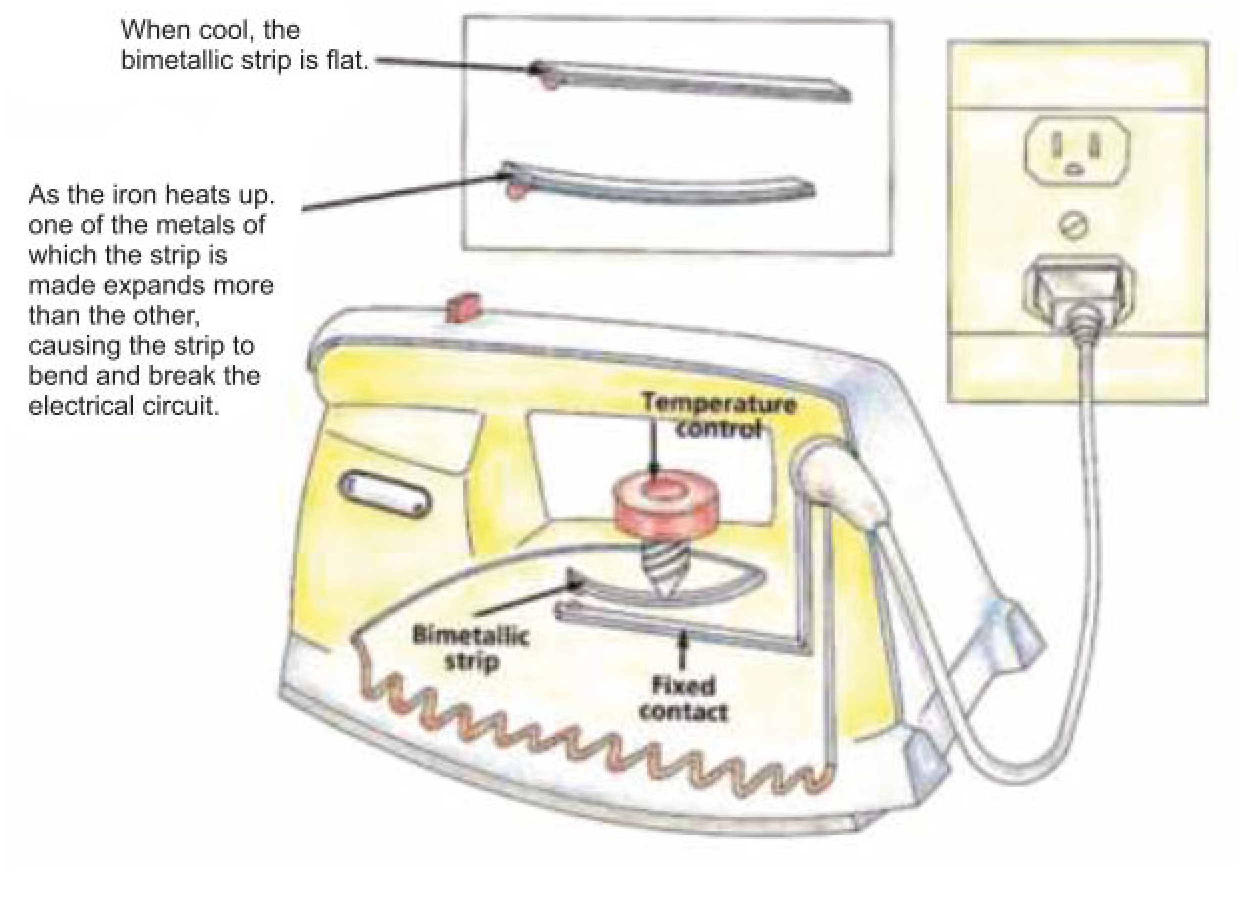


Fig.9.12: Electric Iron

Fire Alarm: Bimetallic strip made up of brass and iron strips is used in fire alarms as shown in Fig 9. 13. When fire breaks out, the bimetallic strip used in the fire alarm becomes hot and bends. Upon bending it touches the contact point of the battery to complete the circuit and the bell connected in the circuit starts to ring to warn of the fire.

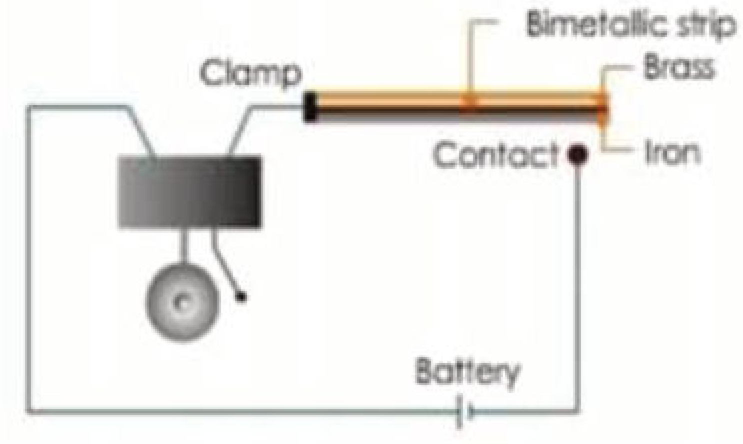


Fig. 9.13 Circuit Diagram of Fire Alarm

Fig. 9.14 Fire Alarm

EFFECTS OF EXPANSION AND CONTRACTION OF SOLIDS IN EVERYDAY LIFE

- Identify the damages caused by expansion and contraction in their surrounding and suggest ways to reduce the damages.
- Investigate the means used by scientist and engineers to overcome the problems of expansion and contraction in everyday life.

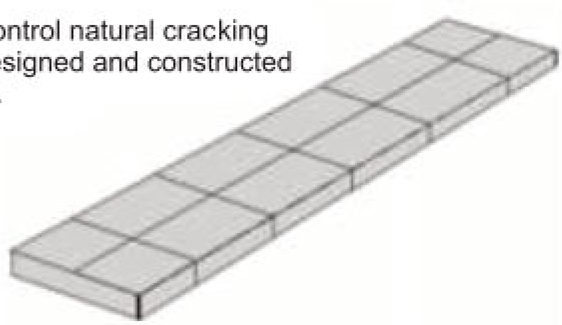
Expansion and contraction of solids create problems. However, scientists and engineers have developed methods to overcome these problems. Some of these are given below.

1.Cracking of Roads and footpaths:

One disadvantage is the cracking of concrete roads and footpaths due to the expansion during hot summer days and contraction during comparatively cold nights. This expansion and contraction make road surface rough. Due to fluctuation in temperature, concrete structure expands or contracts slightly. Temperature change may be caused by environmental conditions or cement dehydration. This simultaneous decrease and increase in size due to change in temperature leads to cracking of structures. In order to overcome this problem, two basic techniques are used:

- **Crack Control Joints:** The most widely used technique to control random cracking in concrete slabs of footpaths and roads is crack control joints. These joints must be established to a depth of the slab thickness. Proper joint spacing and depth are essential to effective control of random cracking.

Joints control natural cracking when designed and constructed properly.



Joints control natural cranking when designed and constructed properly.

Fig. 9.15 Crack control Joints

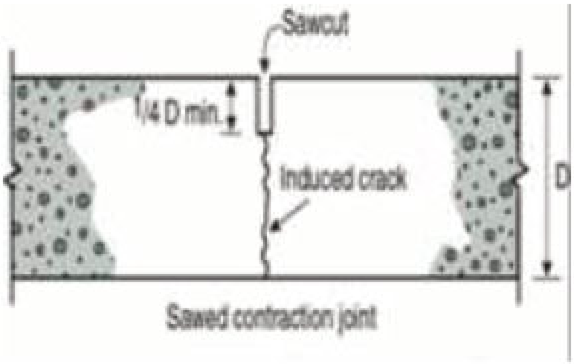


Fig. 9.16 Contraction joint filled with saw

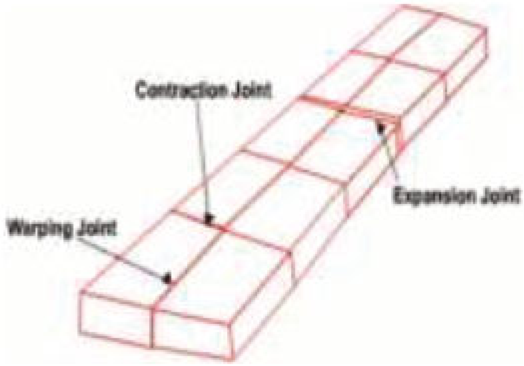


Fig. 9.17 Crack control Joints

- **Steel Reinforcement:** Another technique is to provide steel reinforcement in the slab and structures which holds random cracks tightly. When cracks are held tightly or remain small, the aggregate particles on the faces of crack interlock; thus provide load transfer across the crack.

Figure 9.18 Steel Reinforcement in Building

2. Railway tracks:

Have you observed railway tracks? Why two sections of railway tracks are not welded together? These gaps are meant to control the expansion and contraction during summer and winter seasons. Thus, these gaps prevent railway tracks to de-shape and create hindrance in smooth running of rail. If they are not designed for expansion, then the entire track may bend out of shape during expansion. Rails and bridges expand in hot weather, which can cause them to buckle or break. Railway engineers leave gaps between sections of railway track, which gives the sections room to expand and gives trains their characteristic 'clickety clack' noise when their wheels run over the gaps.

Fig. 9.19 A rail expansion joint, indicated by arrow

3. Expansion of bridges:

Metal and steel structures used in bridges also expand when they heat up, causing fractures in the bridges. Therefore, girders in buildings and bridges are made with gaps at the end. Bridges can be built in sections, connected by expandable joints as shown in Fig 9.20. Predict what will happen if bridges are not designed for the expansion?

Fig. 9.20 A bridge expansion joint

Fig. 9.21 One end of the steel girder bridge is not fixed.

DO YOU KNOW?

1. Oven mitts are used to avoid the extreme of heat of the ovens and pans while cooking.
2. Ski suits prevent skiers to get frost bite by insulating their body from cold.

USES OF EXPANSION AND CONTRACTION OF LIQUIDS

✓ Describe the uses of expansion and contraction of liquids.

Fig. 9.22 Investigate: Why do oil pipelines have so many bends?

Large Bends in Pipes: Water and steam pipes often have a U-bend in them to allow for thermal expansion. In cold weather, liquid or gas in pipes freezes and due to expansion frozen pipes burst. Similarly, when hot liquid or gas flows through pipes they may crack due to expansion or contraction. In order to resolve this problem large bends are given in pipes. The pipes used for transport of petroleum are usually coiled. The coils and curves allow for expansion and contraction so that the pipes may not be damaged.

THERMOMETER

✓ Describe the working of a thermometer.

As you have already explored liquids expand upon heating and contract upon cooling. This property of liquids is used in Thermometer for measuring the temperature. Let us explore how thermometer is constructed and works.

In a thermometer, thermal expansion and contraction of liquid mercury or alcohol is used to measure temperature. You must

Fig. 9.23 Alcohol Thermometer

Fig. 9.24 Mercury Thermometer

have used a mercury thermometer to measure your fever when you are sick. Our normal body temperature is 98.6 degree Fahrenheit (98.6° F). The liquid mercury in a bulb thermometer when heated expands and rises up in the narrow capillary tube.

Mercury thermometer has a thin walled glass bulb filled with mercury and the bulb is attached to a thick-walled glass tube. Liquid expands more than solids. When we place the bulb of thermometer in our mouth under the tongue, mercury of bulb expands and rises in the tube which has linear scale 94° to 108°F (35°C to 42°C).In alcohol thermometers, stained red ethanol is used and temperature can be measured on the scale as alcohol rises through the capillary.

SUMMARY

Sources of Heat Energy

Sun | **Earth** | **Wood** | **Coal** | **Methane** | **Electricity** | **Petroleum** | **Water** | **Oil**

- Heat is the form of energy found due to the movements of atoms and molecules.
- All material objects (solids, liquids and gases) have ability to expand upon heating and contract upon cooling.
- In hot summer days expansion of solids can cause damages. Roads crack because these expand during summer season and contract during the cold season.
- Expansion gaps in concrete roads and railway tracks are used to avoid harmful effects of expansion and contraction.
- One end of iron girders used in bridges is fixed while the other end rests on the rollers.
- Bimetallic strip is used in thermostats. It is made up of two different metal strips welded or riveted together. It bends due to the uneven expansion of two metals.
- Bimetallic strip can be made of Iron and Brass. Brass expands more than the iron strip at the same temperature.
- Large bends and coils are used in pipes carrying hot and cold liquids and gases so that they can expand or contract without cracking.
- Thermal expansion and contraction are used for different purposes such as riveting, fixing the metal tyre over a wheel, and fixing axle into a wheel.
- Water has a peculiar behaviour. It has more density at $4^{\circ}C$ whereas less density at $0^{\circ}C$.

EXERCISE

1. Write answers of the following questions.

- Define thermal expansion. Explain expansion of solids with examples.
- What are the effects of heating and cooling on liquids? Explain with the help of an activity.
- Prove with the help of an experiment that gases expand on heating and contract upon cooling.
- Describe the effects of expansion and contraction of solids. How are these effects overcome?
- How does a bimetallic strip work in thermostat?
- What problems do heat-related expansion cause for bridges or railways?
- Why do telephone wires sag down during summer days?
- Why gaps are left in railway tracks?
- What are the unique freezing properties of water?

2. Choose the correct answer.

i) Which of the following does not make use of the expansion and contraction?

a) An electric fan | b) An electric iron
c) A railway track | d) A thermometer

ii) Which liquid is used in a clinical thermometer?

a) Water | b) Oil
c) Mercury | d) Vinegar

iii) Which of the following substances expands the most; for the same rise in temperature?

a) Air | b) Water
c) Copper | d) Glass

iv) Bimetallic strip is used in

a) Electrical Fan | b) Tape recorder
c) Electric Iron | d) Computer

V) Rivet is used to fasten

a) soft material with hard material
b) soft material with soft material
c) hard material with hard material
d) plastic with soft material

3. Give reasons why?

a) Mercury is used in thermometer.
b) A bimetallic strip is used in electrical gadgets.
c) An iron tyre is heated before fitting it in a wheel.
d) Pipes burst when water freezes inside them.

PROJECT

Making a Bimetal Fire Alarm

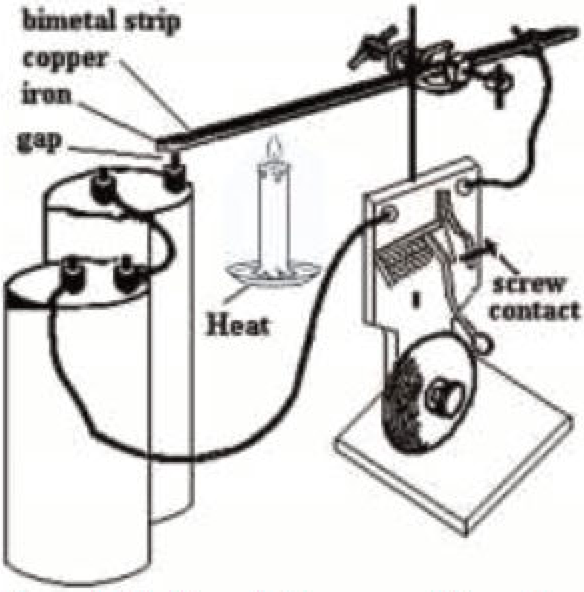


Figure 9.25 Circuit Diagram of Fire Alarm

What I need:

- Insulated copper wire
- Bimetallic strip (Brass and Iron or copper and Iron)
- Iron stand with clamp
- Battery
- Bell
- Candle sprit lamp
- Match box

What to do:

- Connect all the given materials one after another in a loop as shown in the Figure 9.25.
- Connect all the material as shown in the figure/ circuit diagram.
- Light the candle and place it under the free end of bimetallic strip.
- Heat will start bending the bimetallic strip gradually and free end of the bimetallic strip will touch terminal of battery. Circuit will be completed, and the bell will start ringing.